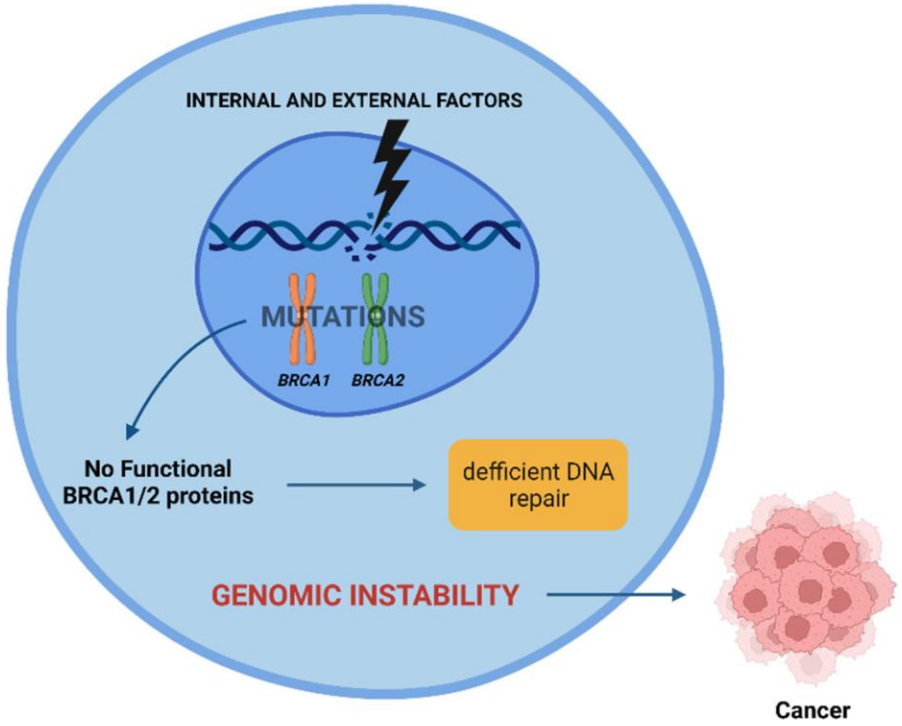
XAI – Final Project

MARYAM MOUNAD

08.10.2026

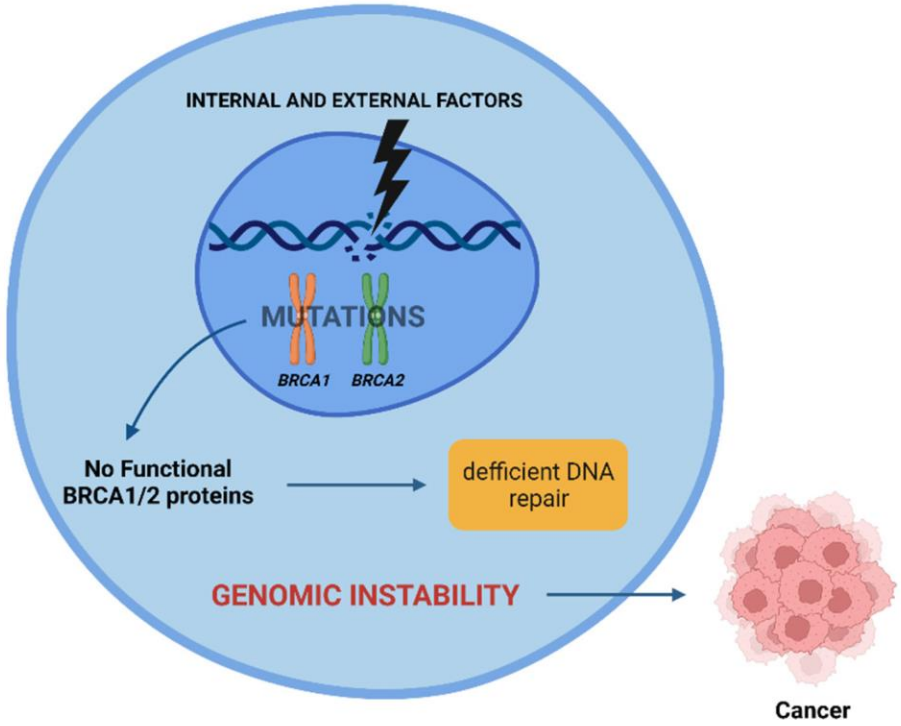
Dataset Selection

- PCBA:
  - Database consisting of biological activities of small molecules generated by high-throughput screening
  - Data Points: 400 000
- Assay ID: BioAssay 624202
- Goal: A screening assay designed to find small molecules that increase (activate) expression of the BRCA1 gene.
- Biological rationale: BRCA1 is a tumor suppressor; restoring/increasing its expression may have therapeutic relevance (e.g., cancers with reduced BRCA1)



Dataset Selection

- PCBA:
- Selected Assay ID: BioAssay 624202
- 365,768 molecules after removing missing values
- Highly imbalanced:
  - Inactive: 361799 Active: 3969

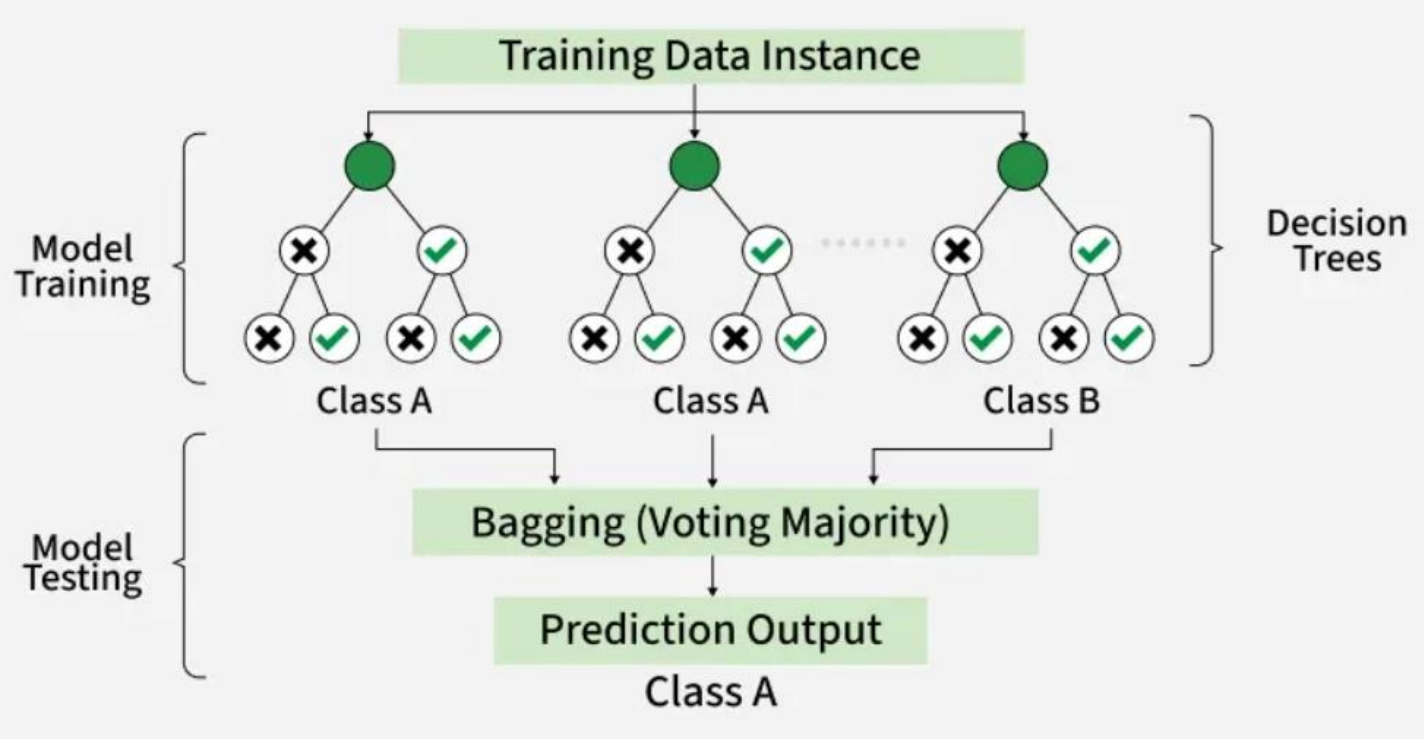


Data Processing

- Converted SMILES to ECFP4 fingerprints
- Split: 80/20 train/test (stratified)
- 73,154 test molecules
- Training set balanced by **undersampling**
  - Before balancing: {0: 289439, 1: 3175} After balancing: {1: 3175, 0: 3175}
- Test set kept at the real imbalance.

Random Forest

- An ensemble machine learning algorithm that operates by building **multiple decision trees** during training and outputting the **average** of the predictions from individual trees for regression tasks, or the **majority vote** for classification tasks.



Random Forest

- Robust on high-dimensional sparse binary features,
- Little preprocessing
- It can be explained with exact TreeSHAP.

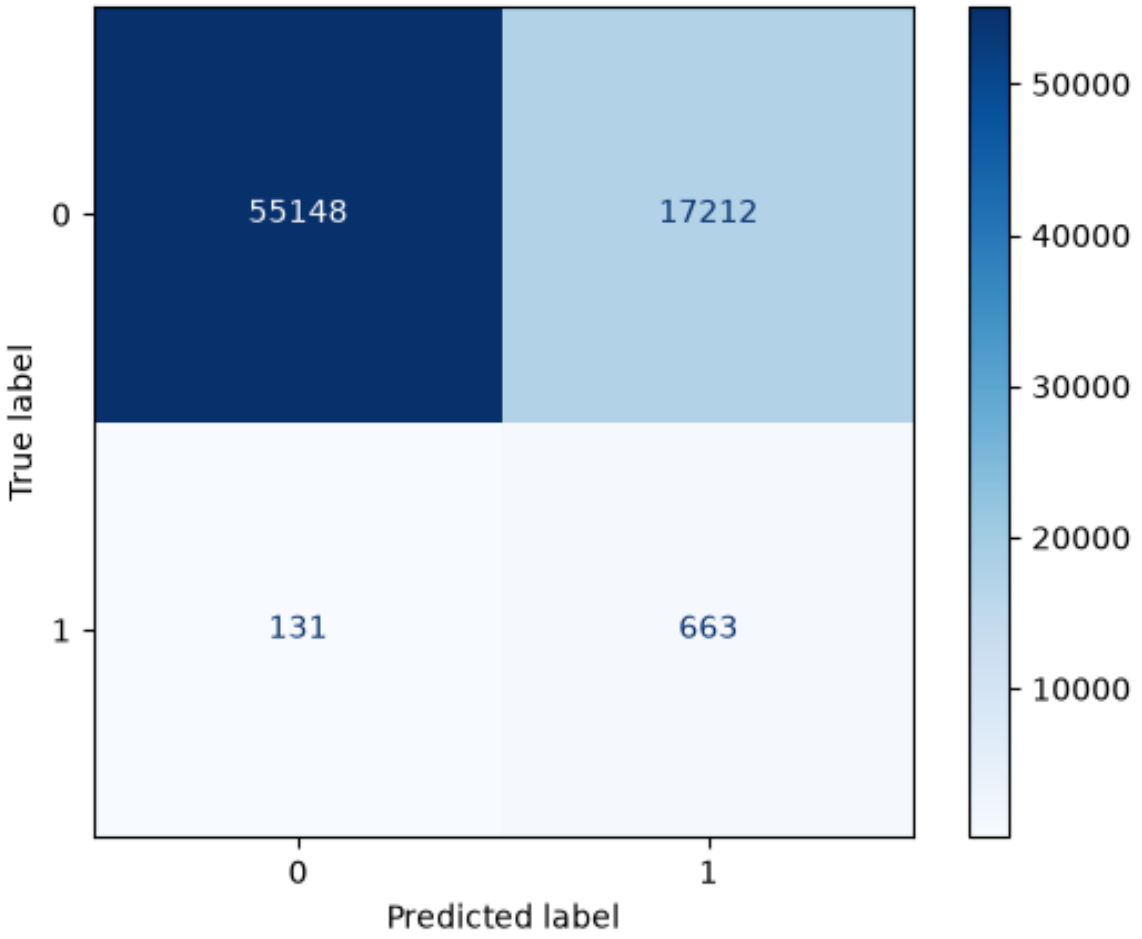
RandomForestClassifier

Parameters

random_state	42
n_estimators	100
criterion	'gini'
max_depth	None
min_samples_split	2
min_samples_leaf	1
min_weight_fraction_leaf	0.0
max_features	'sqrt'
max_leaf_nodes	None
min_impurity_decrease	0.0
bootstrap	True
oob_score	False
n_jobs	None

Model Evaluation

- ROC-AUC Score: 0.8803
- Accuracy Score : 0.7629
- Balanced Accuracy Score: 0.7986



TreeSHAP

- Exact algorithmic (tree-structure traversal)
- Axiomatic (exact Shapley values)
- Model-specific (tree ensembles)
- Local (aggregatable to global)

  - Why: exact and fast for tree ensembles, gives local (per-molecule) and global explanations, and shows a direction (towards active or inactive), unlike built-in importance.

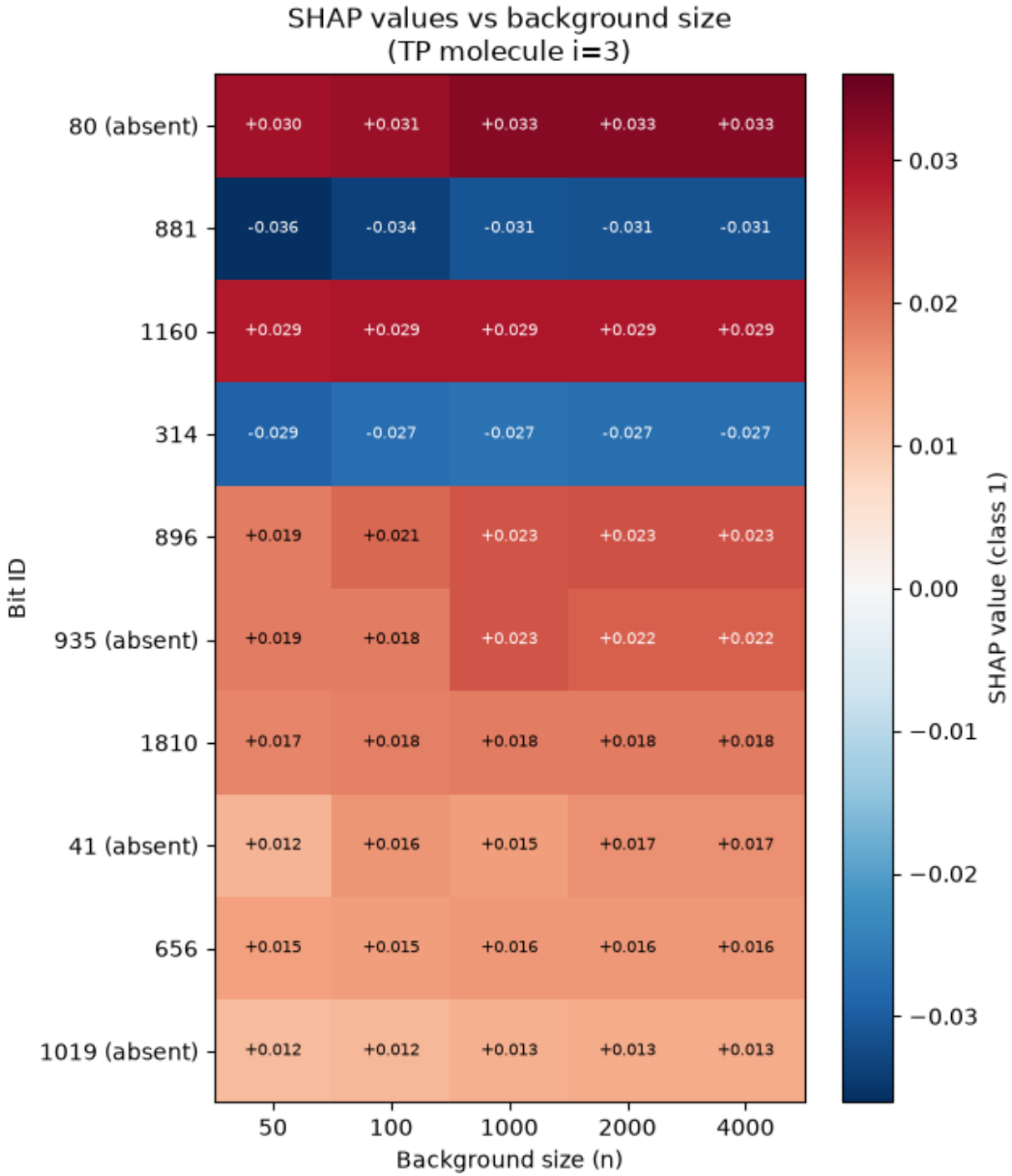
TreeSHAP

- **Interventional TreeSHAP**, probability output, with a background of 1000 molecules from the balanced training set.

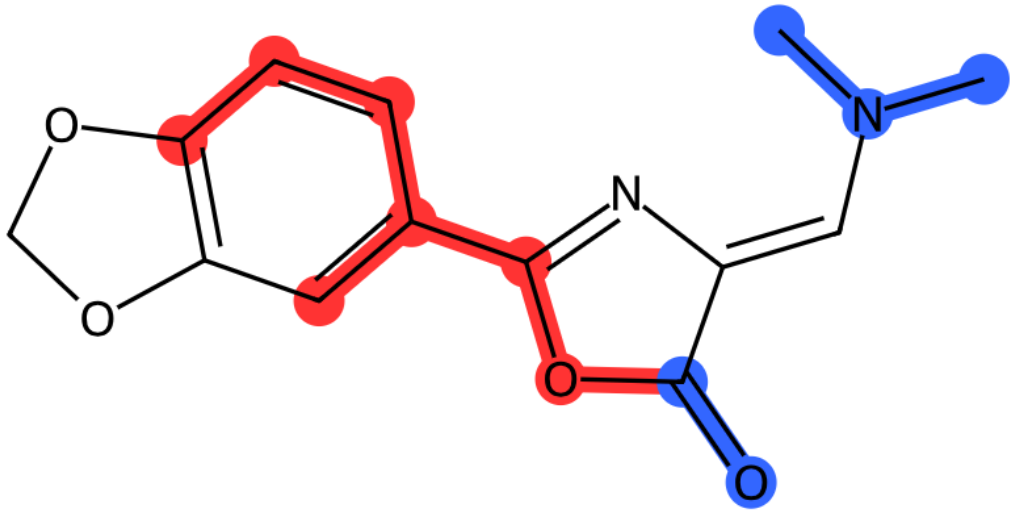
- Path-dependent TreeSHAP was numerically unstable for this model (SHAP sums ~10^{18}), so it wasn't used.

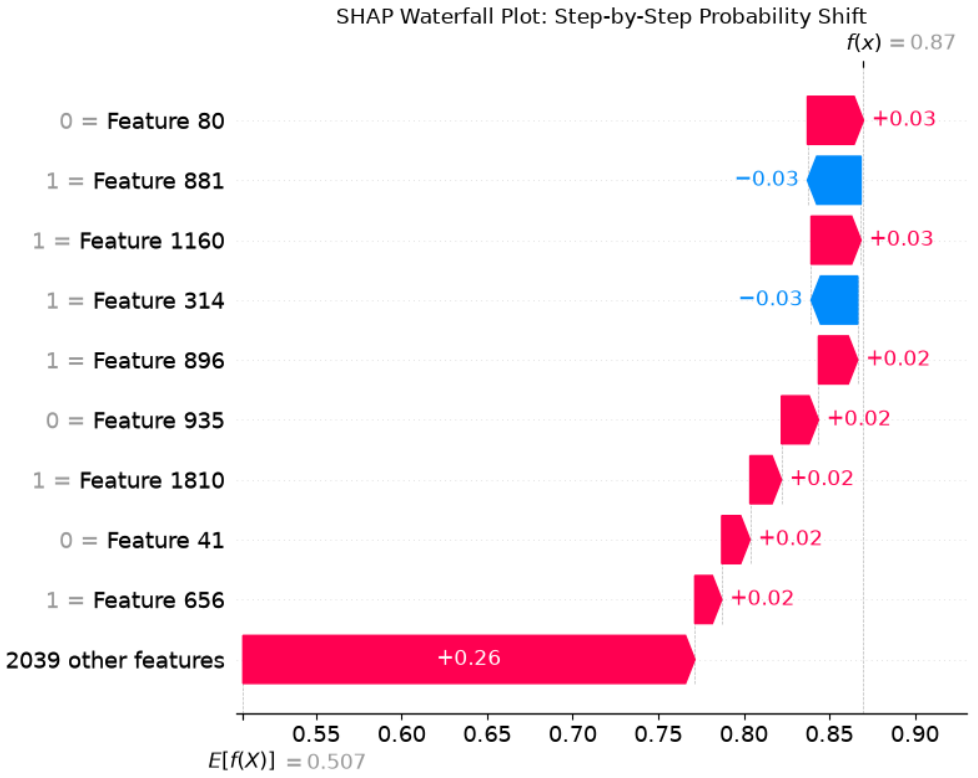
Background-size study

- SHAP values stable from n ≈ 1000.



Example





=== Overlaying Top 3 (+) and Top 3 (-) Present Features ===

Bit ID: 1160 | SHAP: +0.029169 | Active (+)

Bit ID: 896 | SHAP: +0.022930 | Active (+)

Bit ID: 1810 | SHAP: +0.018376 | Active (+)

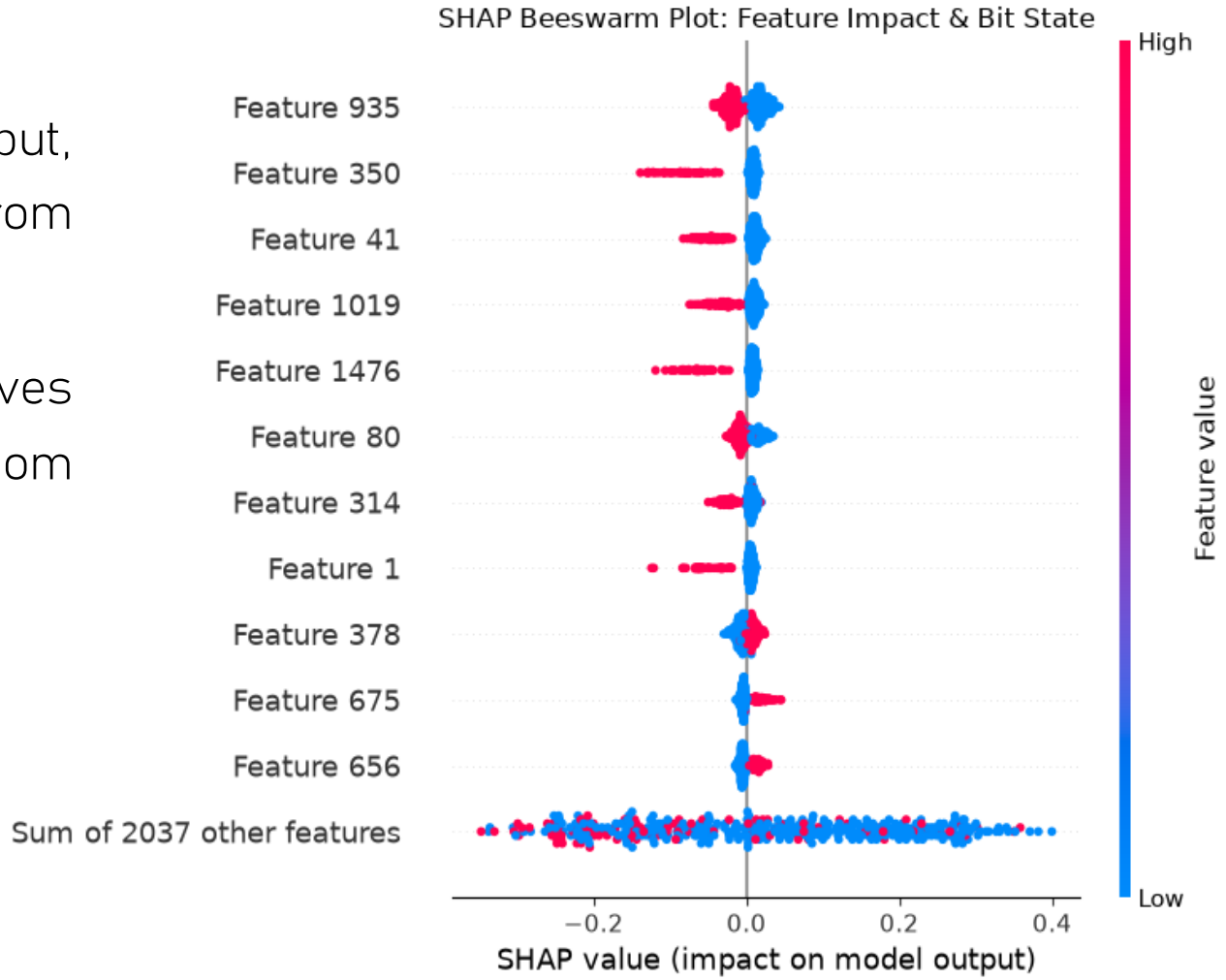
Bit ID: 881 | SHAP: -0.031344 | Non-Active (-)

Bit ID: 314 | SHAP: -0.027233 | Non-Active (-)

Bit ID: 1163 | SHAP: -0.006652 | Non-Active (-)

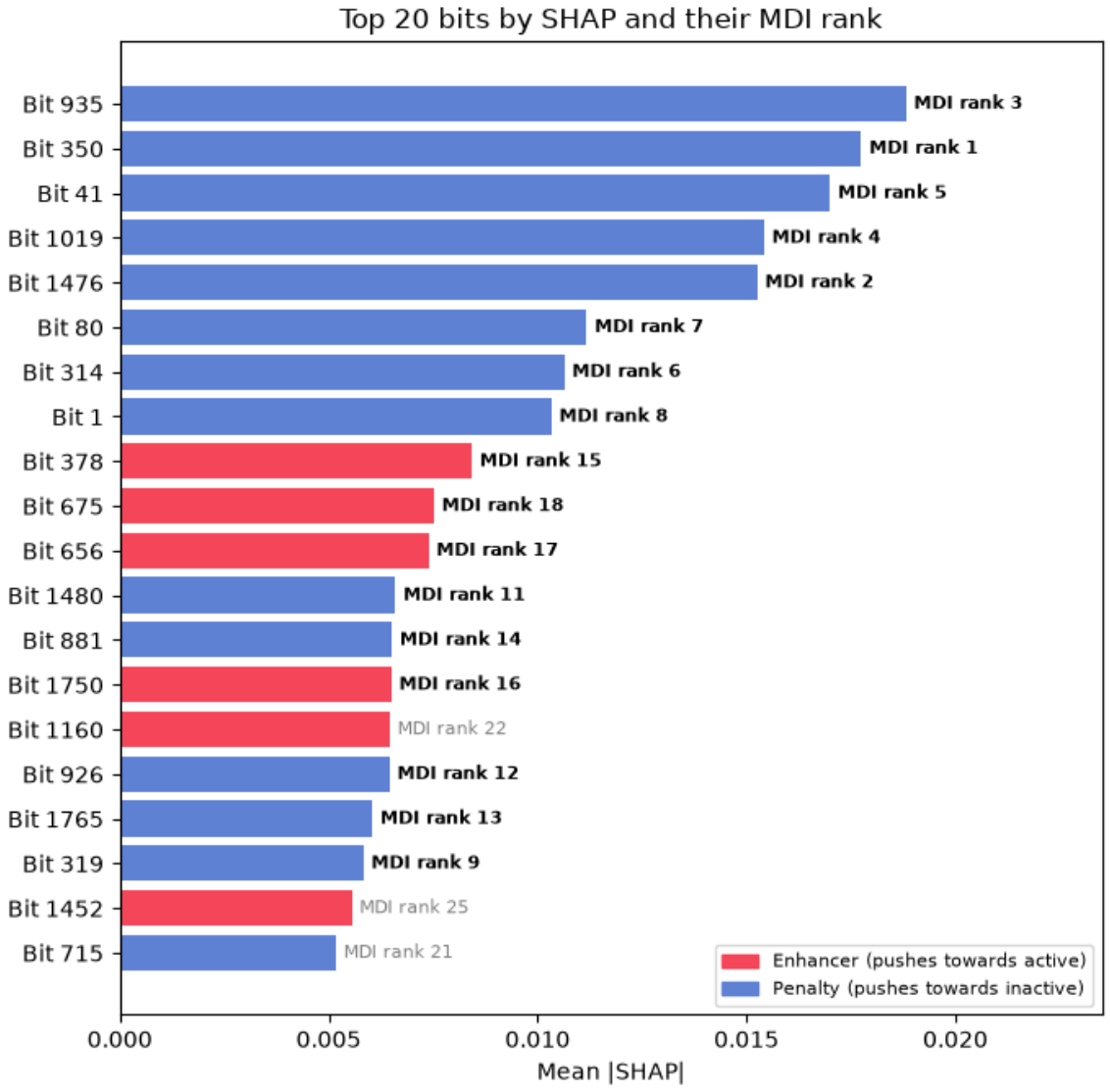
Global Feature Importance

- **Interventional TreeSHAP**, probability output, with a background of 1000 molecules from the balanced training set.
- Used a **stratified test sample** of 200 actives and 200 inactives, because a random sample contained only 6 actives.



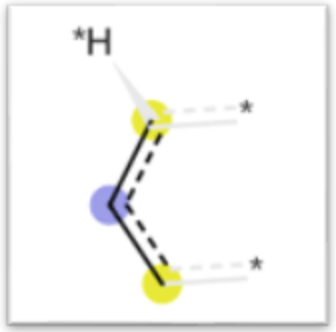
TreeSHAP vs Gini Importance

- 17 of the 20 SHAP top bits are also in the MDI top 20
- The 8 most important bits all push towards inactive when present.
- The model mostly distinguishes actives by the absence of inactive-like substructures, plus a few positive features with smaller effects.



Chemical Interpretation

- Global important features for increased probability of activvity



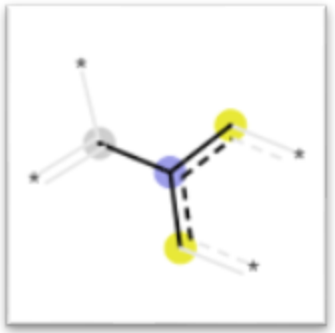
Bit ID 1750



Bit ID 378



Bit ID 656



Bit ID 1160



Bit ID 675

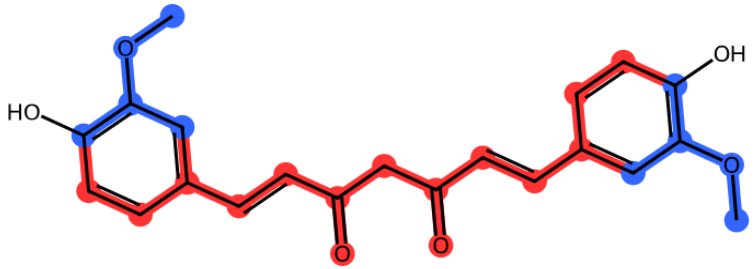
Chemical Interpretation

- **Curcumin** — Epigenetic modifier; demethylates the BRCA1 promoter via TET1 activation and miR-29b induction, restoring BRCA1 mRNA/protein in hypermethylated TNBC cell lines (Al-Yousef et al., 2020).
- **Resveratrol** — Transcriptional transactivator; elevates expressions of BRCA1 and BRCA2 mRNAs in cancer cell lines (Fustier et al., 2003).
- **Indole-3-Carbinol** — Exposure to I3C induces increases in both BRCA1 mRNA and protein across a range of cell types. This upregulation enhances cellular resistance to oxidative stress and DNA-damaging treatments (Centofanti et al., 2023).
- **Genistein** — causes a 4- to 17-fold upregulation of BRCA1 and BRCA2 mRNA and protein in breast and prostate cancer cell models. It demonstrates strong synergy when combined with I3C (Fan et al., 2006).

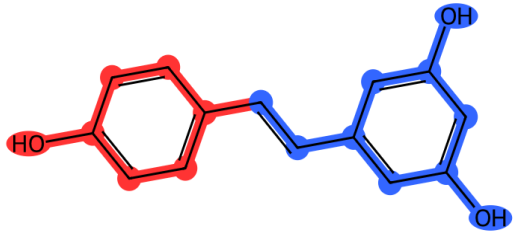
Chemical Interpretation

Top 2 (+) and Top 2 (-) Present Features

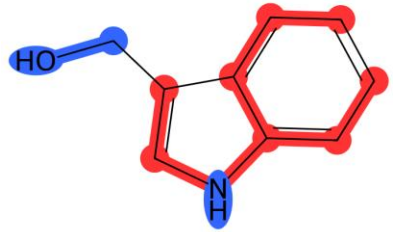
- **Curcumin** — Probability of being active: **0.57**



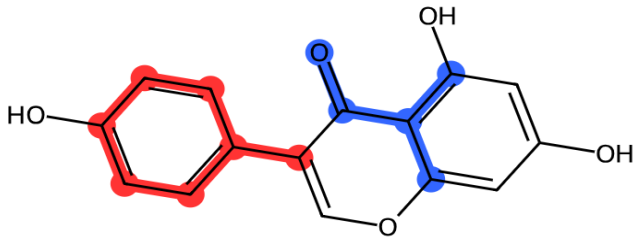
- **Resveratrol** — Probability of being active: **0.84**



- **Indole-3-Carbinol** — Probability of being active: **0.65**



- **Genistein** — Probability of being active: **0.77**



References

Gonçalves, D., Pires, A. S., Marques, I. A., Gomes, I., Sousa, G., Botelho, M. F., & Abrantes, A. M. (2021). An Overview on Radiation Sensitivity in Hereditary Breast and Ovarian Cancer Syndrome. *Cancers*, *14*(13), 3254. https://doi.org/10.3390/cancers14133254

Centofanti, F., Buono, A., Verboni, M., Tomino, C., Lucarini, S., Duranti, A., Pandolfi, P. P., & Novelli, G. (2023). Synthetic Methodologies and Therapeutic Potential of Indole-3-Carbinol (I3C) and Its Derivatives. *Pharmaceuticals*, *16*(2), 240. https://doi.org/10.3390/ph16020240

Fustier, P., Corre, L. L., Chalabi, N., Vissac-Sabatier, C., Communal, Y., Bignon, Y. J., & Bernard-Gallon, D. J. (2003). Resveratrol increases BRCA1 and BRCA2 mRNA expression in breast tumour cell lines. *British Journal of Cancer*, *89*(1), 168. https://doi.org/10.1038/sj.bjc.6600983

Al-Yousef, N., Shinwari, Z., Al-Shahrani, B., Al-Showimi, M., & Al-Moghrabi, N. (2020). Curcumin induces re-expression of BRCA1 and suppression of γ synuclein by modulating DNA promoter methylation in breast cancer cell lines. *Oncology Reports*, *43*(3), 827. https://doi.org/10.3892/or.2020.7473

Fan, S., Meng, Q., Auborn, K., Carter, T., & Rosen, E. M. (2006). BRCA1 and BRCA2 as molecular targets for phytochemicals indole-3-carbinol and genistein in breast and prostate cancer cells. *British Journal of Cancer*, *94*(3), 407. https://doi.org/10.1038/sj.bjc.6602935